

Trust Claims vs. Evidence: An Empirical Study of Documentation Practices in Healthcare AI Models on Hugging Face

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Abstract Healthcare AI systems are increasingly shared as open models on hubs such as Hugging Face, yet the European Union’s AI Act classifies many of them as *high-risk*, demanding transparency about *trustworthiness*, *bias*, and *interpretability*. Whether this transparency is delivered in practice is unclear. To fill this gap, we study 500 healthcare-related model cards on Hugging Face, checking whether claims about trustworthiness, bias, or interpretability are *substantiated* with concrete evidence. Each card was classified three times by an LLM and aggregated via *majority voting* (Fleiss’ $\kappa = 0.678$). We complement this with a control sample of 500 general-purpose models. We find that 90.4% of healthcare model cards make no such claim. The EU AI Act’s entry into force correlates with more clinical-use disclaimers but not more evidence-backed documentation, and an organizational-authorship effect visible in raw counts is not statistically significant once classification noise is accounted for. A multivariate comparison against our control sample shows healthcare models with about 30% lower odds of making any documentation claim than general-purpose models ($p = 0.082$).

Keywords: Healthcare AI · Model documentation · Trustworthy AI · Hugging Face · Empirical study · EU AI Act.

1 Introduction

Healthcare Artificial Intelligence (AI) models are increasingly shared through open model hubs such as Hugging Face [13]. Unlike general-purpose AI, healthcare AI operates within a safety-critical and heavily regulated environment, given that the EU AI Act classifies medical AI systems as *high-risk*, thereby imposing transparency, risk management, and human oversight obligations [1].

Model cards and READMEs are the main channel through which developers communicate a model’s trustworthiness, limitations, and evaluated properties.

Frameworks such as Model Cards [10] and Datasheets for Datasets [6] standardize this disclosure. Such disclosure also represents an entry point for neurosymbolic AI in healthcare [5], since a symbolic, structured claim, such as a predicate tied to a clinical guideline, is far more amenable to automated verification than a free-text assertion that a human or an LLM must interpret.

However, the manner in which disclosure occurs in practice in high-risk domains remains an open question. Prior work on Hugging Face model cards found documentation generally uneven, with sections such as limitations and evaluation frequently empty [8], and dataset, bias, and license information inconsistently reported [11]. These studies code documentation as presence or absence and are not specific to healthcare, so they cannot tell us whether the disclosures that do exist are actually trustworthy or merely present.

This paper provides an empirical analysis of the *claim-evidence gap*, namely whether a card’s assertions about trustworthiness, bias, or interpretability are substantiated with concrete evidence or left unsubstantiated. We apply this to healthcare AI models on Hugging Face and complement it with a control sample of general-purpose models, allowing a direct comparison between a domain that regulation treats as high-risk and the rest of the ecosystem.

We operationalize this into three research questions, spanning regulatory trends, authorship, and a controlled domain comparison. We collected and classified 500 healthcare-related and 500 general-purpose model cards using an LLM-assisted coding scheme that distinguishes *claim* from *evidence*. Each card was classified independently three times by three LLMs instances and aggregated via majority voting [3] (Fleiss’ $\kappa = 0.678$), a reliability check that proves consequential for our findings. The headline result is that over the 90.4% of healthcare model cards make no claim at all about trustworthiness, bias, or interpretability. Where regulation does leave a mark, it does so in the wrong place, since the EU AI Act’s entry into force correlates with more clinical-use disclaimers but not with more evidence-backed documentation. An authorship effect, with organizations appearing to claim more than individuals, is not statistically significant once we account for classification noise, while the comparison against general-purpose models shows healthcare somewhat *less* documented ($p = 0.082$). Taken together, these findings sketch a diagnostic baseline for neurosymbolic AI in healthcare, showing how far current practice remains from the structured, machine-checkable knowledge that neurosymbolic auditing presupposes.

2 Related Work

Documentation frameworks. Mitchell *et al.* [10] introduced *Model Cards*, and Gebru *et al.* [6] *Datasheets for Datasets*, both voluntary frameworks that rely on developers to fill them in thoughtfully; neither enforces that a stated property (*e.g.*, “bias has been mitigated”) is backed by verifiable evidence. Hugging Face’s model-card template is inspired by this work but remains optional and unevenly filled.

Both frameworks describe what documentation should look like; the next question is what it actually looks like in practice.

Empirical documentation studies. Liang *et al.* [8] analyzed 32,111 Hugging Face model cards, finding documentation generally uneven and popularity positively correlated with completeness. Pepe *et al.* [11] examined how models document training datasets, bias, and licenses, finding pervasive gaps. Both studies code documentation as presence or absence, *i.e.*, whether a topic is mentioned at all, on the general model population. Neither distinguishes a card claiming a property from substantiating it, and neither focuses on healthcare.

A recurring neurosymbolic AI motivation is that neural-model behavior becomes auditable only once checkable against explicit, structured knowledge rather than prose [5]; our claim-evidence scheme imposes exactly this kind of lightweight symbolic structure on free-text documentation.

≡ Novelty and Positioning.

We differ from prior work along three axes. (1) We focus on *healthcare*, the domain the EU AI Act designates high-risk [1]. (2) We code a *claim-evidence gap*, checking whether an assertion is backed up with evidence. (3) Instead of comparing indirectly against numbers from studies with different sampling and coding, we build a *self-collected, methodologically identical control group* for an internally valid domain comparison (RQ3), using prior literature only as motivation.

3 Research Method

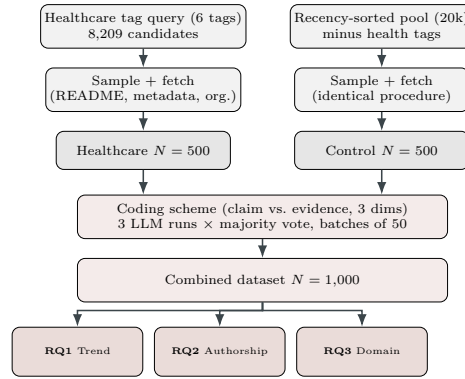


Figure 1: Research method overview. Two identical pipelines collect a healthcare and a control sample, both classified (3 runs, majority vote) and merged to answer RQ1–RQ3.

Driven by the Goal Question Metric paradigm [2], our *goal* is to characterize the extent to which trustworthiness, bias, and interpretability claims in healthcare AI model cards are substantiated with concrete evidence, with respect to regulatory timing, popularity, authorship, and domain, from the *viewpoint* of researchers, platform maintainers, and regulators, in the context of the Hugging Face ecosystem, using a self-collected general-purpose sample as a basis of comparison. From this goal we derived three research questions, each targeting a different explanation for the gap.

Q RQ₁. Regulatory trend.

Has documentation of trustworthiness, bias, and interpretability improved over time, particularly around EU AI Act milestones?

The goal of RQ₁ is to test whether regulation is the driver: if the Act’s transparency obligations work as intended, documentation should improve specifically around its milestones rather than following a generic upward trend.

Q RQ₂. Authorship.

Do organization-published models show a smaller claim-evidence gap than individually-published ones?

The goal of RQ₂ is to test whether authorship compensates where regulation does not, distinguishing a resource/process gap at the individual level from a broader norm even well-resourced actors do not follow.

Q RQ₃. Domain comparison.

Is healthcare documented differently from the rest of the ecosystem, controlling for popularity and authorship?

The goal of RQ₃ is to test the central premise of risk-based regulation itself, namely whether healthcare is documented differently from the rest of the ecosystem; this requires an external control group rather than an indirect comparison with differently collected prior statistics.

Figure 1 outlines the method, detailed below.

Data Collection. We queried the HF Hub API for models tagged `medical`, `clinical`, `healthcare`, `biomedical`, `radiology`, or `pathology` (8,209 unique candidates), randomly sampled and fetched each candidate’s raw `README.md` plus metadata (author, downloads, likes, creation date, pipeline tag; near-empty cards <50 characters were discarded and replaced), and checked author namespaces against the HF Organizations API for `is_organization`. This yielded 500 valid healthcare models.

Coding Scheme. Once collected, each card required a scheme capable of distinguishing a genuine assertion from a vague mention of a topic. We targeted three dimensions, namely trustworthiness/safety, bias/fairness, and interpretability. For each we coded a binary `claim` field, requiring an explicit positive assertion, and a 3-valued `evidence` field (a named method, metric, test, benchmark, or citation backing the claim). We additionally coded `clinical_use_disclaimer` (yes/no) and `interpretability_method_named` (*e.g.*, SHAP, Grad-CAM, or none), and derived `gap_score` (0–3), the number of dimensions with `claim=yes` and `evidence=no`.

Classification and Reliability. Manually applying this scheme to 1,000 long and heterogeneous cards was not feasible, so we automated the process. Cards were split into 10 batches of 50 and classified using **Claude (Sonnet 5)**⁴ via Claude Code’s agentic orchestration, with zero-shot [12] and rubric-based [14] prompt, and every batch was classified independently three times (3,000 classifications total) and majority-voted per field. Ties on `evidence` resolve to “partial”; `interpretability_method_named` counts as present if at least 2 of 3 runs agree. We measured Fleiss’ κ [4] per field, pooled across healthcare and control, ranging from 0.525 to 0.920 with an overall average $\kappa = 0.678$ (substantial agreement). The two rare-category bias fields sit at the low end despite $\geq 98.6\%$ raw agreement, a known artifact when prevalence is near zero rather than a sign of poor classification.

Control Group. A numeric comparison against prior work’s published statistics [8,11] would be invalid, since their coding and sampling differ from ours. We instead built a **self-collected control group** of 20,000 most-recently-created HF models, excluding health-tagged ones, sampled and processed identically, yielding **500 valid general-purpose models**.

Analysis. With both datasets classified, we tested each RQ as follows. For RQ1, we applied a chi-square test [15] on documentation rates before and after the EU AI Act’s entry into force (1 Aug. 2024). For RQ2, we applied a Mann-Whitney U test [9] on `gap_score` and chi-square tests on `any_claim` and `disclaimer`, broken down by authorship. For RQ3, on the combined dataset ($N=1,000$), we fit a multivariate logistic regression [7] $\text{any_claim} \sim \text{domain} + \log(1+\text{downloads}) + \text{is_organization}$ and an analogous OLS on `gap_score`, controlling for popularity alongside domain and authorship.

4 Results

Unless stated, all numbers use majority-voted labels across the 3 classification runs, only **9.6%** of the 500 healthcare model cards make a claim on any of the three dimensions, meaning 90.4% state nothing at all (Table 1). Bias/fairness claims are nearly absent (0.8%, 4 models); a specific interpretability method is named in only 1.2% of cards (6/500). The strict claim-without-evidence gap

⁴ <https://www.anthropic.com/claude/sonnet>

occurs in only 1.6% of cards (8/500). A clinical-use disclaimer is present in 18.0%; 29.8% of models are organization-authored.

Table 1: Claim/evidence rates per dimension (healthcare, majority-vote, $N=500$).

Dimension	% claiming	% with full evidence
Trustworthiness/safety	7.8%	43.6% ($n=39$)
Bias/fairness	0.8%	75.0% ($n=4$)
Interpretability	1.4%	42.9% ($n=7$)

Overstating trustworthiness without evidence is rare. The dominant issue is that most models simply say nothing, as for 90.4% of them there is no assertion at all to formalize as a checkable symbolic predicate. The remaining three research questions ask whether this silence is temporary, specific to individual developers, or specific to healthcare itself.

4.1 RQ1 – Regulatory Trend

The most direct candidate explanation is regulation, since if the EU AI Act is changing behavior, documentation should improve around its milestones. We compare pre/post EU AI Act entry into force (1 Aug. 2024, Table 2).

Table 2: Documentation rates before/after EU AI Act entry into force.

Period	N	Any claim	Disclaimer
Pre (before 2024-08)	80	6.3%	8.8%
Post (from 2024-08)	420	10.2%	19.8%

The any-claim increase is directionally positive but **not significant** ($\chi^2=0.815$, $p=0.367$); the disclaimer increase **is significant** ($\chi^2=4.800$, $p=0.0285$). Regulation, therefore, altered what developers write, without affecting the part that would actually benefit a downstream user.

Summary

Substantiated claims did not increase after the EU AI Act’s entry into force, but disclaimers did. This asymmetry survives the reliability check and looks like *compliance by disclaimer* rather than genuine transparency improvement.

4.2 RQ2 – Organizational vs. Individual Authorship

If regulation is not closing the gap, perhaps institutions are, given that organizations have review processes and reputational exposure that individual contributors typically lack.

Table 3: Documentation by authorship type (healthcare, majority-vote).

Group	N	Any claim	Gap_score	Disclaimer
Organization	149	12.8%	0.013	6.7%
Individual	351	8.3%	0.017	22.8%

Organizations do make claims more often in absolute terms, but the gap is **not statistically significant** ($\chi^2=1.940$, $p=0.164$). Gap_score does not differ either way (Mann-Whitney $p=0.766$). What does hold up is the opposite of what we expected, namely that individuals disclaim $3\times$ more often than organizations.

Summary

The organizational-authorship effect on any-claim rate is not statistically significant, suggesting the raw difference is partly a classification-noise artifact. Only the disclaimer asymmetry, with individuals disclosing more, is significant.

4.3 RQ3 – Healthcare vs. Control (Domain Comparison)

Neither regulation nor institutional authorship closes the gap on its own. The last remaining question is whether healthcare needs correcting at all, since documentation might simply be just as sparse everywhere, with nothing domain-specific left to explain. Comparing healthcare ($N=500$) to the self-collected control sample ($N=500$) yields an any-claim rate of 9.6% for healthcare versus 13.0% for control, organizational authorship of 29.8% versus 20.4%, a disclaimer rate of 18.0% versus 7.4%, and identical median downloads of 9.0 for both groups. In unadjusted terms, healthcare is *lower*, $\chi^2=2.554$, $p=0.110$. Table 4 reports the multivariate regression, which also controls for popularity alongside domain and authorship.

Table 4: Logistic regression predicting any_claim from domain, log(downloads), and organizational authorship ($N=1,000$).

Predictor	OR	95% CI	p
Healthcare domain	0.70	[0.47, 1.05]	0.082
log(1+downloads)	0.99	[0.90, 1.09]	0.888
Organization-authored	1.26	[0.81, 1.96]	0.298

Controlling for popularity and authorship, healthcare shows **30% lower odds** of any claim than control ($p=0.082$, just short of the conventional 0.05 threshold). Popularity itself is not a significant predictor here either, consistent with its null role throughout our data. The analogous OLS on `gap_score` shows no significant predictor (domain $p=0.445$; downloads $p=0.818$; org $p=0.712$). Healthcare is therefore not merely as sparse as the rest of the ecosystem; if anything, it is sparser, in precisely the domain that regulation has deemed to deserve greater scrutiny, not less.

Summary

Healthcare trends lower than general AI (OR= 0.70, $p=0.082$), just short of significance, pointing toward the high-risk domain being *less* documented, the opposite of what risk-based regulation would predict.

5 Implications

The reliability-checked comparison points toward healthcare being somewhat *less* documented than the broader ecosystem.

For **regulators**, the EU AI Act singles out healthcare as high-risk with heightened transparency obligations [1], yet our RQ3 finding, healthcare trending lower in any-claim odds ($p = 0.082$), suggests this has not translated into more careful documentation. Paired with RQ1’s pattern of more disclaimers without more evidence, this suggests category-level regulation may need concrete platform-level requirements to actually change practice.

For **platform maintainers**, Hugging Face’s card template already has limitations/bias sections, yet these are overwhelmingly unused or generic, mirroring general-population findings [8,11]. Mandatory evidence fields for self-tagged high-risk models, or prompts to justify claims at upload, could help; a longer-term, neurosymbolic option is machine-checkable symbolic assertions verified by a rule engine, turning a claim into a specification [5].

For **practitioners**, the rarity of the strict gap (1.6%) is not reassuring, as it reflects silence rather than substantiation. Roughly 1 in 10 models offer any signal on trustworthiness, bias, or interpretability, and just over 1 in 100 name a concrete method. A model card’s presence alone should not be taken as evidence of adequate risk assessment.

For **researchers**, our claim-vs-evidence scheme and self-collected control group offer a template beyond presence/absence coding. Extensions include tracking documentation debt along model-lineage chains, restricting to models with clinical-decision `pipeline_tags`, and replicating RQ3 on larger samples or other platforms.

6 Threats to Validity

Construct validity. Classification was performed by a single LLM (Claude Sonnet 5), zero-shot rubric-based, with no human validation; our $\kappa = 0.678$ reliability check measures agreement between repeated LLM passes, not ground truth, so a manually-coded sample remains future work. Temperature and top- p were set by Claude Code’s orchestration layer rather than by us, which may shift absolute rates even after voting, though this affects both samples equally and mainly threatens absolute rather than relative comparisons. The two rare-category bias fields sit at the low end of the κ range despite $\geq 98.6\%$ raw agreement, a near-zero-prevalence artifact rather than poor classification.

Internal validity. Majority voting reduces but does not eliminate correlated errors across runs of the same model. RQ1’s disclaimer-rate association with the EU AI Act’s entry into force is correlational; other trends may coincide with it.

External validity. Both samples ($N=500$ each) are Hugging-Face-specific and may not generalize elsewhere; tag-based retrieval may miss or over-include borderline models. The control group’s “high-stakes-use disclaimer” generalizes the healthcare “clinical-use disclaimer” field and is not perfectly equivalent.

Conclusion validity. Tests were not corrected for multiple comparisons, so RQ3’s $p=0.082$ should be read with caution, and subgroup sizes are modest ($n=80$ pre-regulation in RQ1). RQ2 and RQ3’s borderline results are sensitive to the specific classification and aggregation procedure, underscoring real uncertainty in our estimates. Card text truncation (3,000 characters) could omit evidence in unusually long cards.

7 Conclusions

This paper studied documentation practices for healthcare AI models on Hugging Face, distinguishing a *claim* about trustworthiness, bias, or interpretability from its *substantiation*. Each card was classified three times and majority-voted to guard against classification noise. The dominant failure mode is silence, as 90.4% of 500 healthcare model cards make no claim at all. The EU AI Act’s entry into force correlates with more disclaimers but no more evidence, a *compliance-by-disclaimer* pattern that held up under the reliability check, while popularity shows no effect at all. An organizational-authorship effect visible in raw counts was not statistically significant once majority-voted, showing it was partly noise. A self-collected control sample of 500 general-purpose models revealed healthcare with about 30% lower odds of any claim than control.

Future work should add a manually-coded validation sample, extend to models with clinical-decision pipeline tags, and study documentation-debt propagation along model-lineage chains. This underscores the value of repeated classification and majority voting when using LLMs as annotators, since individual passes can be noisy. Closing the gap between regulatory intent and practice will likely require platform-level, machine-checkable documentation requirements, extending neurosymbolic integration from the models themselves to how their properties are documented.

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References

1. Regulation (eu) 2024/1689 of the european parliament and of the council of 13 june 2024 laying down harmonised rules on artificial intelligence (artificial intelligence act). Official Journal of the European Union (2024)
2. Basili, V.R., Caldiera, G., Rombach, H.D.: The goal question metric approach. In: Encyclopedia of Software Engineering, pp. 528–532. Wiley (1994)
3. Chen, L., Davis, J., Hanin, B., Bailis, P., Stoica, I., Zaharia, M., Zou, J.: Are more llm calls all you need? towards the scaling properties of compound ai systems. *Advances in Neural Information Processing Systems* **37**, 45767–45790 (2024)
4. Fleiss, J.L.: Measuring nominal scale agreement among many raters. *Psychological Bulletin* **76**(5), 378–382 (1971). <https://doi.org/10.1037/h0031619>
5. d’Avila Garcez, A., Lamb, L.C.: Neurosymbolic ai: The 3rd wave (2020)
6. Gebru, T., Morgenstern, J., Vecchione, B., Vaughan, J.W., Wallach, H., Daumé III, H., Crawford, K.: Datasheets for datasets. *Communications of the ACM* **64**(12), 86–92 (2021). <https://doi.org/10.1145/3458723>
7. Glonek, G.F., McCullagh, P.: Multivariate logistic models. *Journal of the royal statistical society: series b (Methodological)* **57**(3), 533–546 (1995)
8. Liang, W., Rajani, N., Yang, X., Ozoani, E., Wu, E., Chen, Y., Smith, D.S., Zou, J.: Systematic analysis of 32,111 ai model cards characterizes documentation practice in ai. *Nature Machine Intelligence* **6** (2024). <https://doi.org/10.1038/s42256-024-00857-z>
9. MacFarland, T.W., Yates, J.M.: Mann–whitney u test. In: *Introduction to non-parametric statistics for the biological sciences using R*, pp. 103–132. Springer (2016)
10. Mitchell, M., Wu, S., Zaldivar, A., Barnes, P., Vasserman, L., Hutchinson, B., Spitzer, E., Raji, I.D., Gebru, T.: Model cards for model reporting. In: *Proceedings of the Conference on Fairness, Accountability, and Transparency (FAT* ’19)*. pp. 220–229. ACM (2019). <https://doi.org/10.1145/3287560.3287596>
11. Pepe, F., Nardone, V., Mastropaolo, A., Canfora, G., Bavota, G., Di Penta, M.: How do hugging face models document datasets, bias, and licenses? an empirical study. In: *Proceedings of the 32nd IEEE/ACM International Conference on Program Comprehension (ICPC 2024)*. pp. 370–381. ACM (2024). <https://doi.org/10.1145/3643916.3644412>
12. Pourpanah, F., Abdar, M., Luo, Y., Zhou, X., Wang, R., Lim, C.P., Wang, X.Z., Wu, Q.J.: A review of generalized zero-shot learning methods. *IEEE transactions on pattern analysis and machine intelligence* **45**(4), 4051–4070 (2022)
13. Riva, M., Parigi, T.L., Ungaro, F., Massimino, L.: Hugging face’s impact on medical applications of artificial intelligence (2024)
14. Sharma, M., Zhang, C.B.C., Bandi, C., Wang, C., Aich, A., Nghiem, H., Rabbani, T., Htet, Y., Jang, B., Basu, S., et al.: Researchrubrics: A benchmark of prompts and rubrics for evaluating deep research agents. In: *International Conference on Learning Representations*. vol. 2026, pp. 90447–90472 (2026)
15. Tallarida, R.J., Murray, R.B.: Chi-square test. In: *Manual of pharmacologic calculations: with computer programs*, pp. 140–142. Springer (1987)